

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



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NGHIÊN CỨU GIẢI PHÁP TỐI ƯU
CẤU TRÚC CỦA ĐỘI HÌNH ROBOT
PHỐI HỢP ĐẨY VẬT

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SOA and Related Works

1.1 Inertial Parameters Estimation

To the best of our knowledge, there has been no comprehensive review dedicated specifically to center-of-mass (CoM) estimation for manipulated objects. Existing surveys primarily focus on the identification of the complete set of inertial parameters rather than CoM estimation itself. A survey of Schedlinsk [1] presents a literature review of existing methodologies on how to estimate these parameters in mechanical engineering, whereas Mavrakis and Stolkin [2] is specific to robotics. According to Mavrakis and Stolkin's taxonomy, the methods for estimating inertial parameters can be categorized into three main groups by degree of interaction — purely visual, exploratory, and fixed-object.

The first visual methods relied on the assumption of a known and uniform object density in order to simplify the estimation problem by reducing it to the computation of the object's volume. In [3] the object is represented as a collection of equally sized voxels with identical masses, from which the center of mass is determined by computing the moment of inertia matrix. One of the most widely recognized approaches is presented in [4], the author decomposed a rigid body into polyhedral components with uniform density and computed the mass, and inertia tensor of each component before combining them to obtain the inertial properties of the complete object. The computation was achieved by converting the required three-dimensional volume integrals into two-dimensional surface integrals and one-dimensional line integrals using the divergence theorem, projection, and Green's theorem. However, while visual perception can accurately capture an object's shape and volume, the underlying density distribution is generally unknown, may vary throughout the object, and cannot be directly observed solely from visual information.

Exploratory and fixed-object methods, on the other hand, involve physical interaction with the object to perform the estimation task. These interactive techniques are commonly formulated based on the Newton–Euler equations of motion, which establish a mathematical relationship between the forces/torques (wrenches) applied to an object and its resulting motion. Properties like the center of mass (CoM), mass and the inertia tensor are treated as constant parameters within these equations and are subsequently estimated from the resulting system of equations, typically using least-squares techniques. In fixed object methods [5, 6, 7], the unknown object is rigidly connected to the robotic system, during the estimation process, either through a firm grasp or by being directly attached to the end-effector. Excitation trajectories are then executed by the robotic system to retrieve the necessary data from wrist sensors. The approach proposed in [8] employed a robotic arm to perform three distinct trajectories to distinguish the individual effects of mass, CoM, and the inertia tensor. The fundamental principle of this method can also be applied to different robotic platforms. In [9], the authors use a UAV equipped with two customised grippers to grasp the object and estimated the payload mass, CoM, and inertia from the UAV dynamics and measured motion, while [10] extended this concept to a hexacopter equipped with a 2-DOF manipulator, enabling simultaneous payload estimation and adaptive motion control. Although this method can achieve the most accurate full three-dimensional inertial properties under controlled conditions, its requirements for a secure grasp and specialized force/torque sensors makes it less versatile than exploratory methods in unconstrained or hazardous scenarios.

Exploratory methods exploit the measurements by utilizing more basic interaction such as such as poking, pushing or tilting. As mentioned, early studies of this category similarly estimates the unknown properties by using the model-based approach. Yu et al. [11] employed a two-fingered robot manipulator equipped with force sensors to estimate an object’s mass, 2D CoM, rotational inertia, and motion friction parameters from multiple pushing motions using a least-squares approach. For CoM estimation, they introduced C.M. lines, which are parallel to the object’s straight-motion direction and pass through its CoM. These lines were identified from fingertip force and motion measurements under different pushing directions, with their intersection determining the CoM. Franchi et al.[12] developed a decentralized approach for estimating an object’s mass, 2D CoM, and rotational inertia using multiple mobile robots. Information on the applied forces and contact geometry is shared among the robots through a consensus algorithm, enabling distributed

estimation of the unknown inertial properties. The method was subsequently extended in [13] to rely solely on noisy end-effector velocity measurements.

Another emerging line of research focuses on CoM estimation without requiring an explicit dynamic model. A number of studies have been built upon Mason’s Voting Theorem [14], which establishes a geometric relationship between the pushing direction and the resulting rotational motion of a planar object, and reducing the search space through successive bisection. By additionally incorporating modified binary search algorithm [15], Hyland et al. [16] proposed iterative pushing method using a holonomic mobile robot and external motion capture to derive the CoM location from multiple lines on which the CoM lies, called lines of action (LoA). This works is later extended in [17] into an infrastructure-free approach by utilizing onboard force and vision sensors. Uemura et al. [18] presented a dynamic control based method utilizing a nonlinear feedback control law to continuously adjusting the finger’s motion to suppress the object’s rotation, achieving pure translational motion along the identified line of action (LoA).

A growing body of work has explored data-driven approaches that learn CoM-related information directly from observed interaction data. Gao et al. [19] introduce a force-sensor-free exploratory framework that integrates high-speed robotic pushing with Mason’s Voting Theorem and recurrent neural networks to estimate the CoM of previously unseen objects with non-uniform density. In [20], the authors introduces a data-driven framework that estimates an object’s 2D inertial parameters by training a Multi-Output Regression Random Forest on physical features extracted from a single robotic push. By combining analytical pushing mechanics with machine learning, the method achieves robust and accurate localization across various surface types without the need for specialized hardware. These methods enable rapid estimation with fewer physical interactions. However, their reliance on prior training data may limit their generalization to previously unseen objects.

Methodology

2.1 Methodology

2.1.1 Formulation

The system consists of N robots manipulating an object with polygonal shape. Each robot can exert a normal force f_i within a limited range : $0 < f_i < f_{max}$ at its contact point C_i . The net force and torque that all robots exert on the object are describe as:

$$\mathbf{F} = \sum_{i=1}^N \mathbf{F}_i + \mathbf{F}_f \quad (2.1)$$

$$\tau = \sum_{i=1}^N (\mathbf{p}_i - \mathbf{p}_C)^{\perp T} \mathbf{F}_i + \tau_f \quad (2.2)$$

where $(\cdot)^{\perp}$ denotes the perpendicular operator, defined as $\mathbf{v}^{\perp} = Q\mathbf{v}$, with $Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

$$\tau = \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i - \mathbf{p}_C^{\perp T} \sum_{i=1}^N \mathbf{F}_i + \tau_f \quad (2.3)$$

$$\tau = \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i + \tau_f \quad (2.4)$$

$${}^B\mathbf{p}_C = R^T(\theta) \left[{}^W\mathbf{p}_r - {}^W\mathbf{p}_o + \frac{1}{\omega} ({}^W\dot{\mathbf{p}}_r)^{\perp} \right] \quad (2.5)$$

2.1.2 Formation synthesis

$$\tau_{\max} = \max_{0 \leq f \leq f_{\max}} \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^N \mathbf{F}_i = \mathbf{0}. \quad (2.6)$$

$$\tau_{\min} = \min_{0 \leq f \leq f_{\max}} \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^N \mathbf{F}_i = \mathbf{0}. \quad (2.7)$$

$$J_r = \min(|\tau_{\max} - \tau_d|, |\tau_{\min} - \tau_d|). \quad (2.8)$$

$$\begin{aligned} \min_{\mathbf{p} \in \mathbb{R}_{\geq 0}^N} \quad & J(\mathbf{p}) = -J_r(\mathbf{p}) \\ \text{s.t.} \quad & \exists \tilde{\mathbf{f}} \in \mathcal{F} \quad \text{s.t.} \quad \sum_{i=1}^N \mathbf{F}_i = \mathbf{0}, \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i = \tau_d, \\ & \text{frD}(p_i, p_j) \geq d_{\min}, \quad \forall i \neq j. \end{aligned} \quad (2.9)$$

2.1.3 Force-based control

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